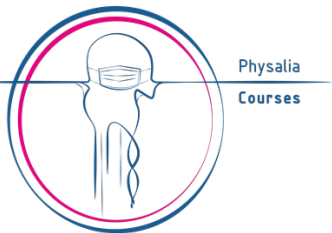
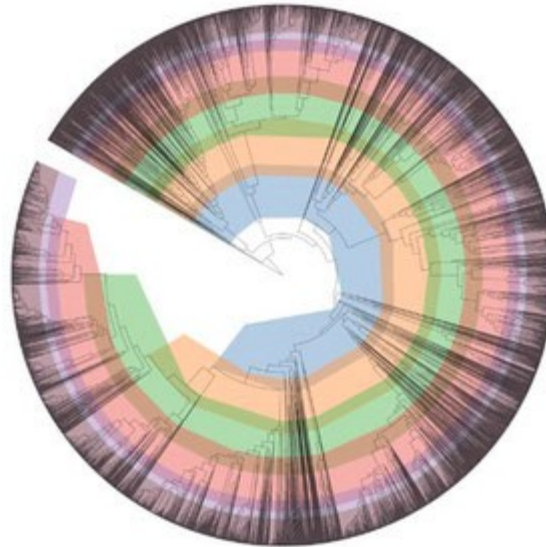


ENVIRONMENTAL METAGENOMICS

Physalia course, online, 13-17 October 2025

Data pre-processing: quality control and adapter removal

Nikolay Oskolkov, Group Leader of Metabolic Research Group at LIOS, Riga, Latvia
Samuel Aroney, Postdoctoral Research Fellow, Queensland University of Technology



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NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)

FASTQ File

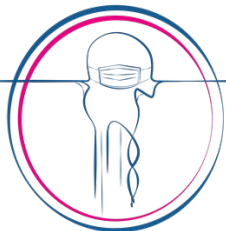
Example (files can be gigabytes in size!)

```
@K00233:37:HGHLBBXX:3:1101:2646:1121 1:N:0:NACGCATC+NGCTGGTG
NCGCATGAGCCGCCTGTATCAGGCGCTGATCGGGCCGGGCATTGCAGTTGGGATAGATCGGGGGAGCACACGTCTG
+
#A7F<<GG<JFJFJJJJJJJFFJJJJJJJAFFJFJJJJJJJFJAFFFJAJFJJ<FJJJJJFFF<FFA--FFFJJJJJ
@K00233:37:HGHLBBXX:3:1101:4655:1121 1:N:0:NACGCATC+NGCTGGTG
NATGCATGACAGGAGGTGAGGGCATTTTCCAGATTTTCAGGCTGCGACCTTGAGCATCTTTCGCCGCTTCCAGCAC
+
#GG-<FFFF7JFF7JJJJJFJJ<JJJJJA7FJJJJJJJFF<JFF<J7-<FJJJJFJFFJJJGGGGFFJJ--AJAJJ
```

```
@ <read id, e.g. machine ID, location on flowcell> <extra metadata>
  <DNA sequence; Note: N = base couldn't be called!>
+ <a separator>
  <base quality scores for each nucleotide in sequence>
```

Quality score:

```
!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJ
0.2.....26...31.....41
```



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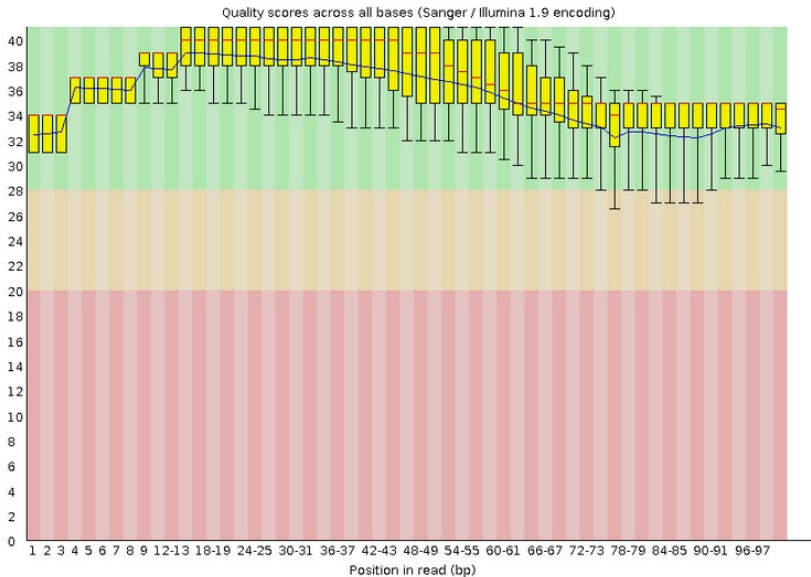
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	G69146_pe_1.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	8997912
Sequences flagged as poor quality	0
Sequence length	77-101
%GC	49

Per base sequence quality



Quality Scores

A quality score (or Q-score) expresses an error probability. In particular, it serves as a convenient and compact way to communicate very small error probabilities.

Given an assertion, A, the quality score, Q(A), expresses the probability that A is not true, P(~A), according to the relationship:

$$Q(A) = -10 \log_{10}(P(\sim A))$$

where P(~A) is the estimated probability of an assertion A being wrong.

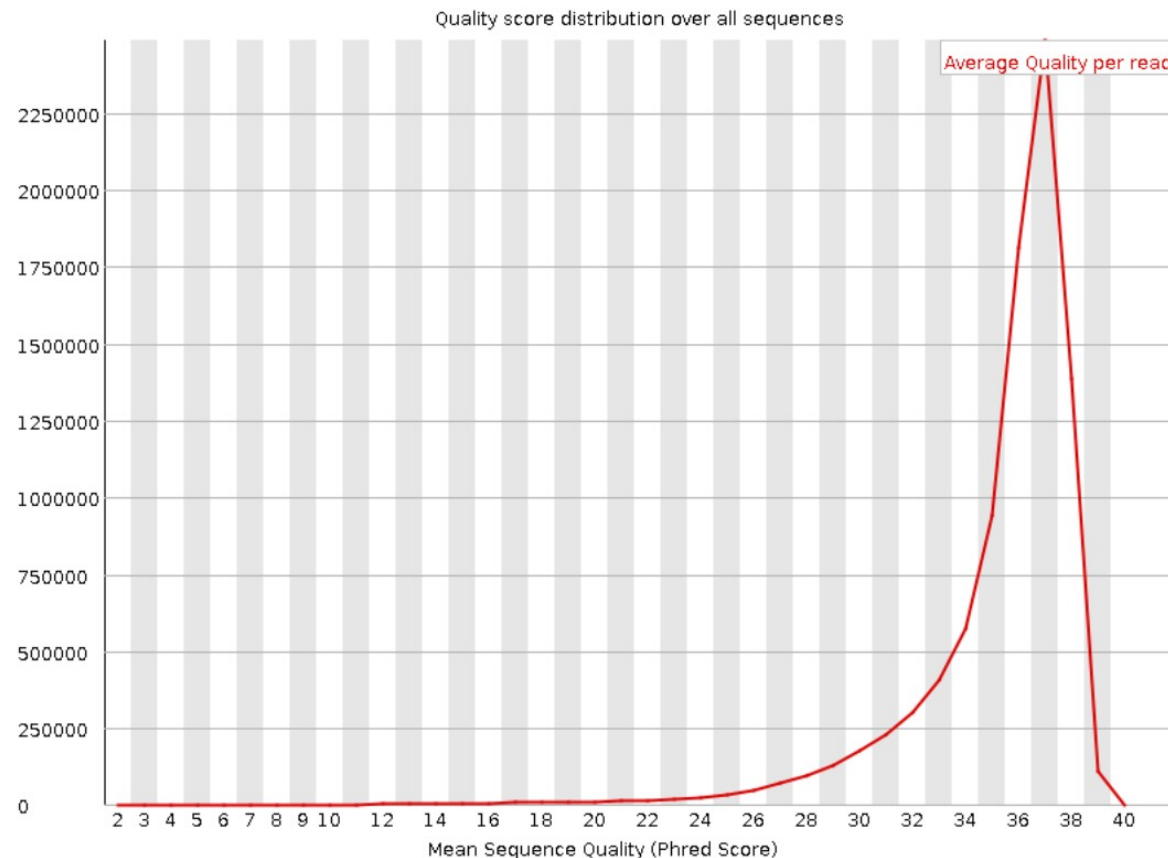
The relationship between the quality score and error probability is demonstrated with the following table:

Quality score, Q(A)	Error probability, P(~A)
10	0.1
20	0.01
30	0.001

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ! [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ! [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)

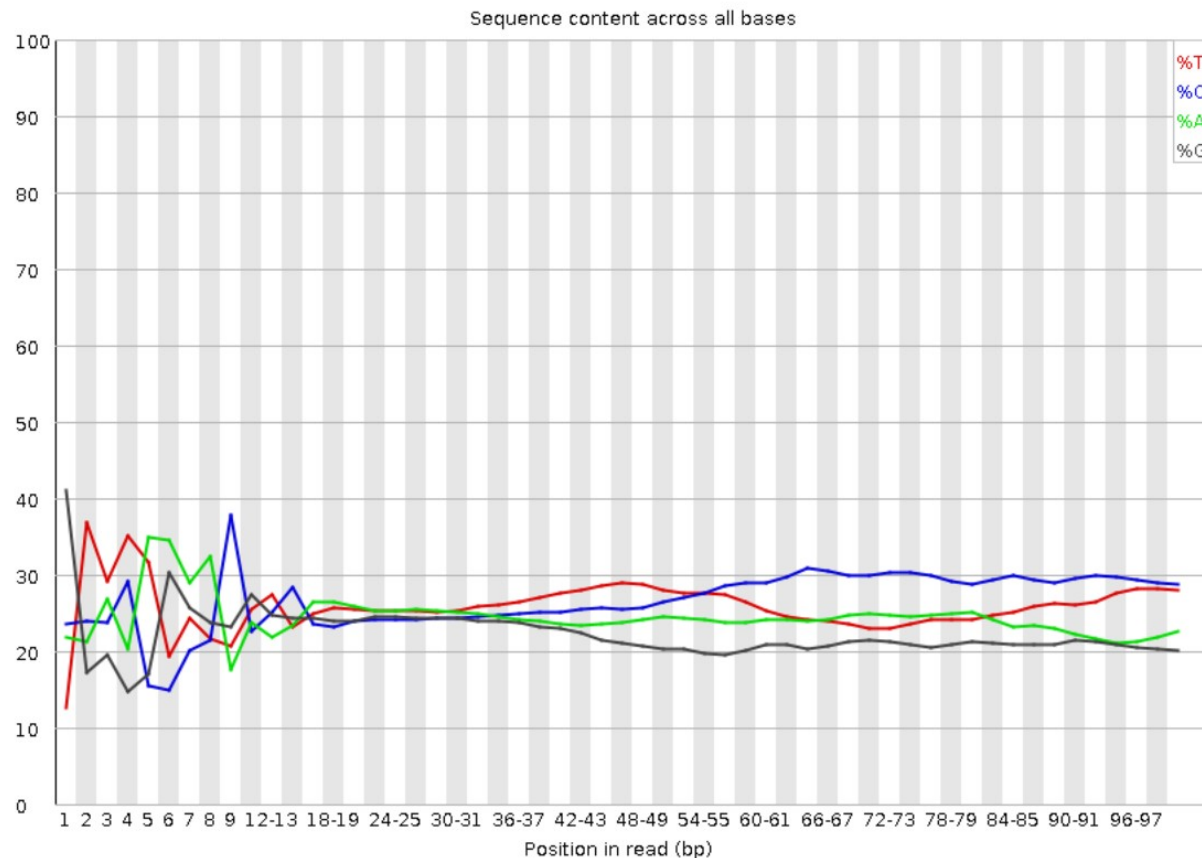
✓ Per sequence quality scores



Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ! [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ! [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)

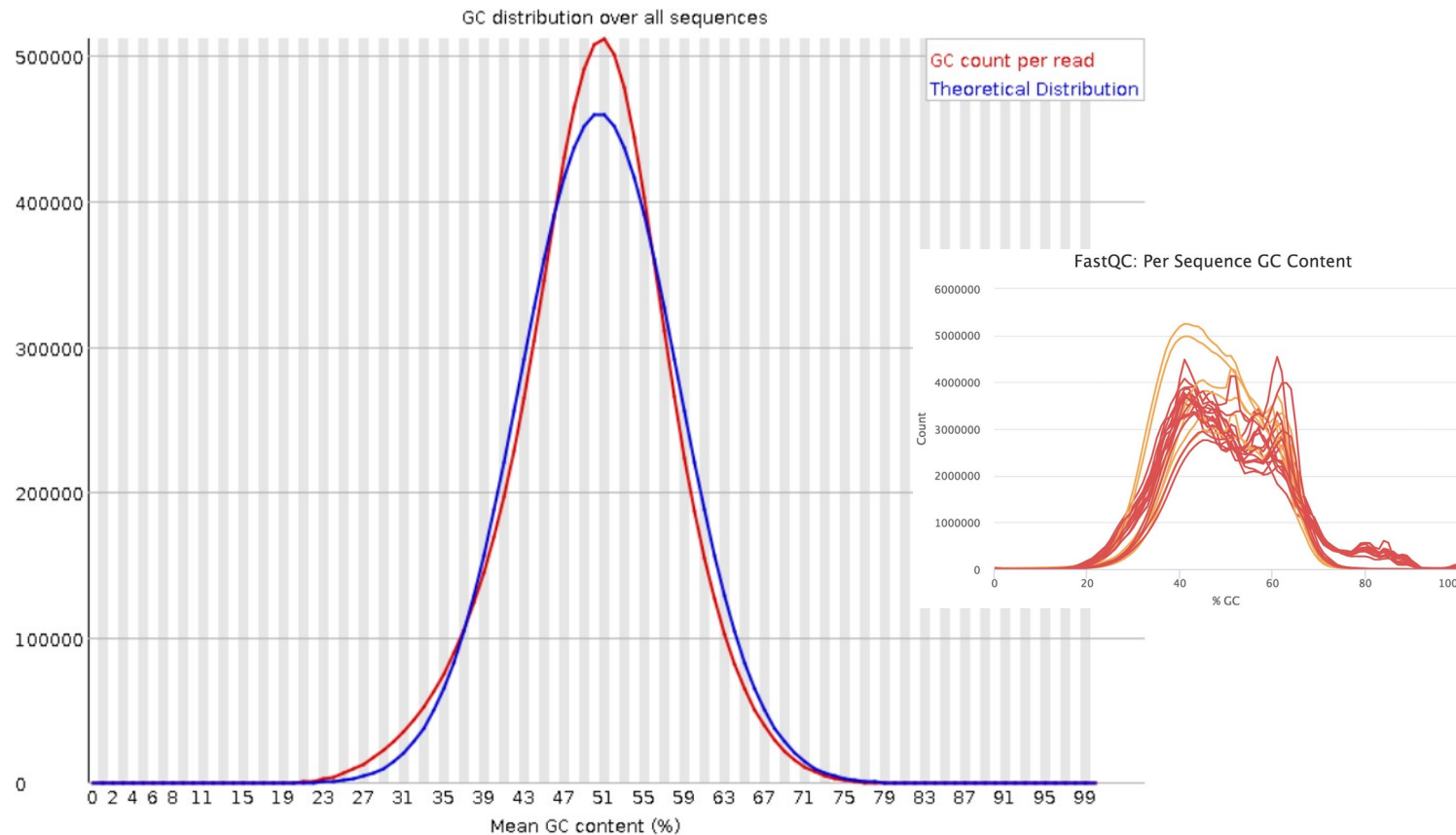
! Per base sequence content



Summary

- ✓ Basic Statistics
- ✓ Per base sequence quality
- ✓ Per tile sequence quality
- ✓ Per sequence quality scores
- ! Per base sequence content
- ✓ Per sequence GC content
- ✓ Per base N content
- ! Sequence Length Distribution
- ✓ Sequence Duplication Levels
- ✓ Overrepresented sequences
- ✗ Adapter Content

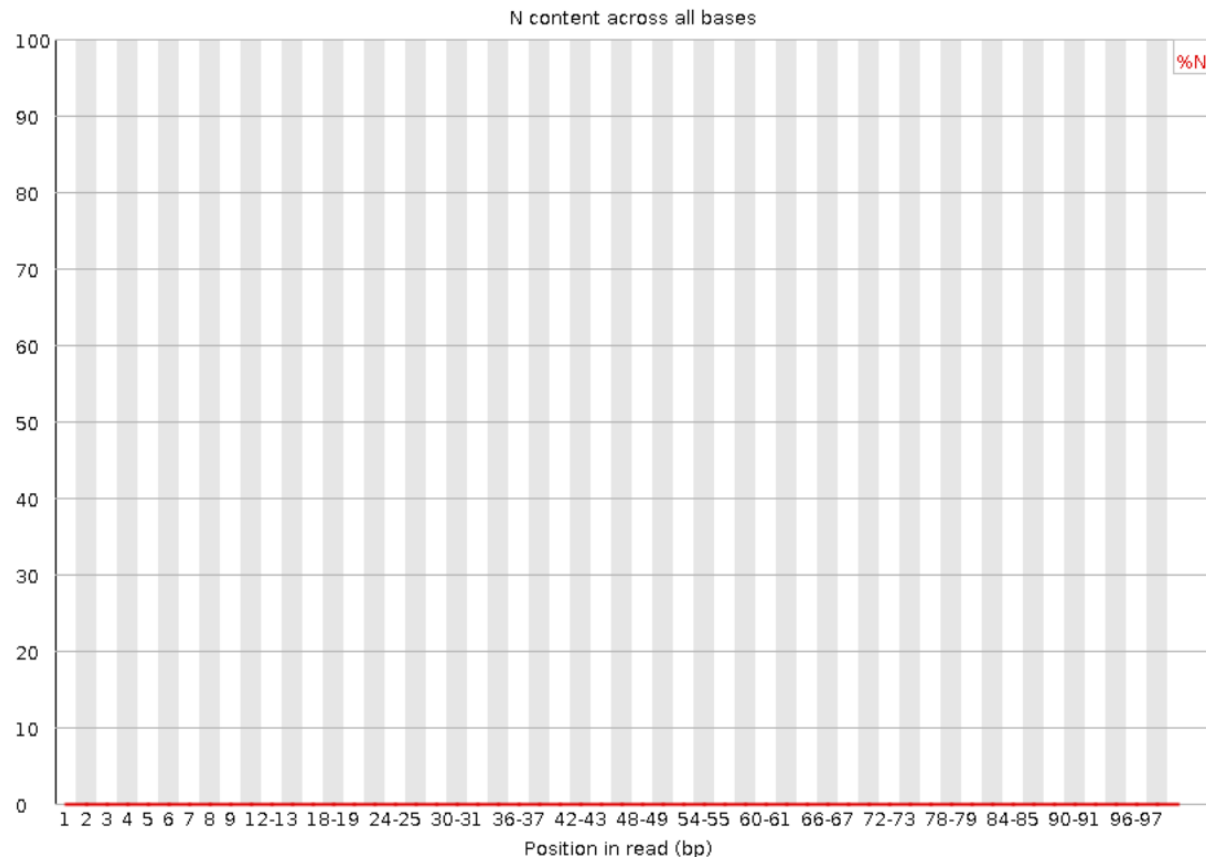
✓ Per sequence GC content



Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ! [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ! [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)

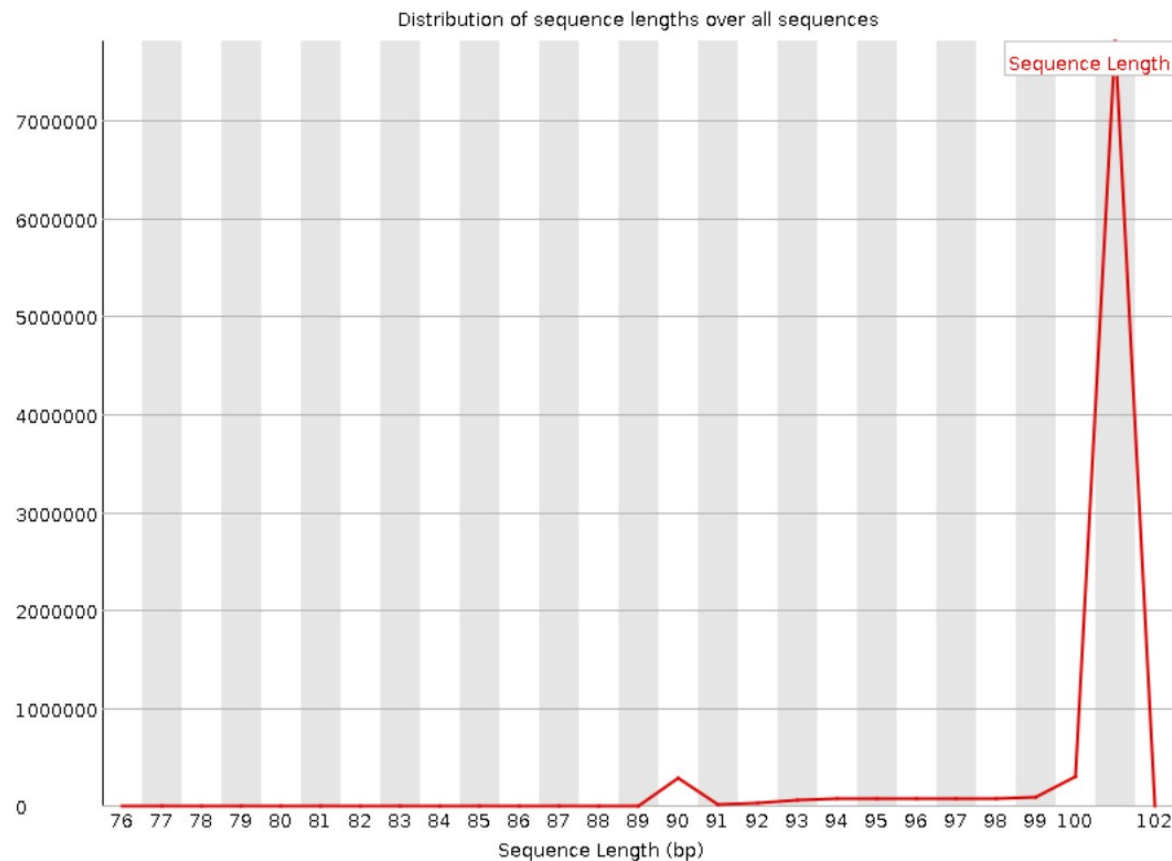
✓ Per base N content



Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ! [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ! [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)

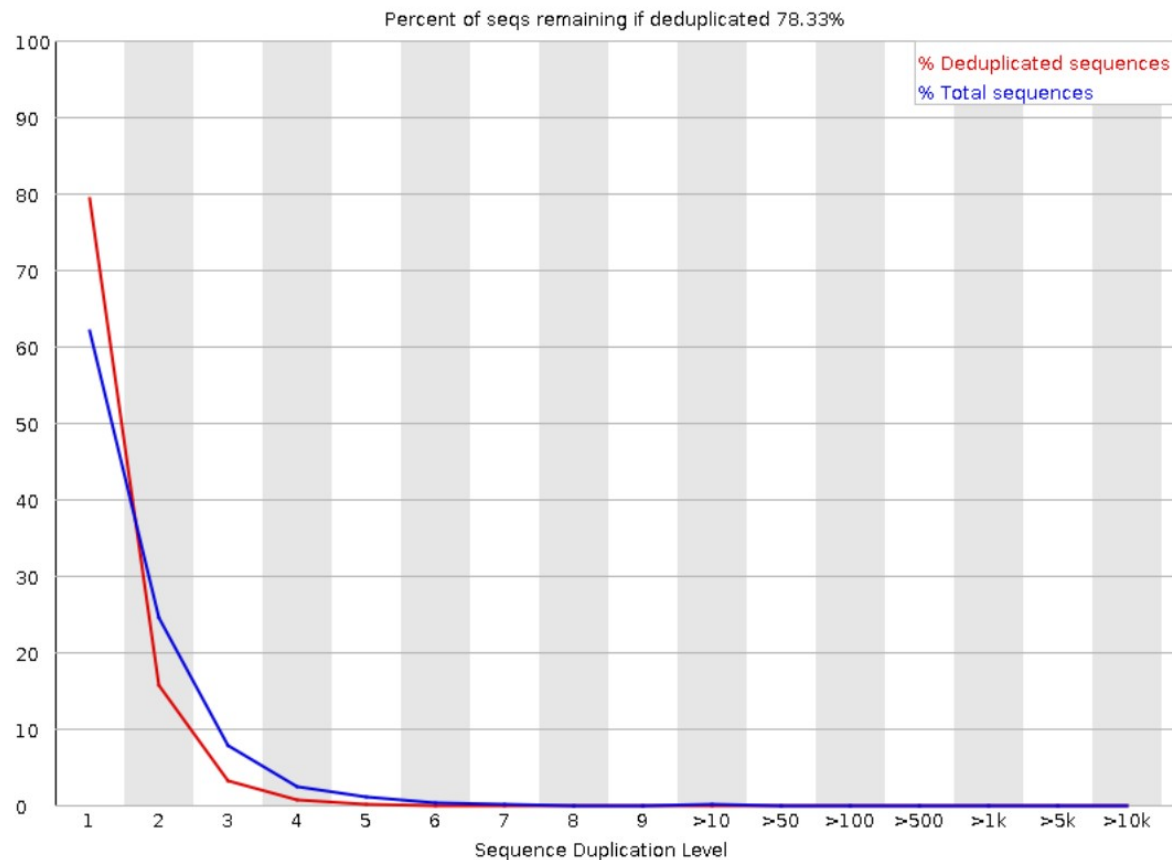
! Sequence Length Distribution



Summary

- ✓ Basic Statistics
- ✓ Per base sequence quality
- ✓ Per tile sequence quality
- ✓ Per sequence quality scores
- ! Per base sequence content
- ✓ Per sequence GC content
- ✓ Per base N content
- ! Sequence Length Distribution
- ✓ Sequence Duplication Levels
- ✓ Overrepresented sequences
- ✗ Adapter Content

✓ Sequence Duplication Levels



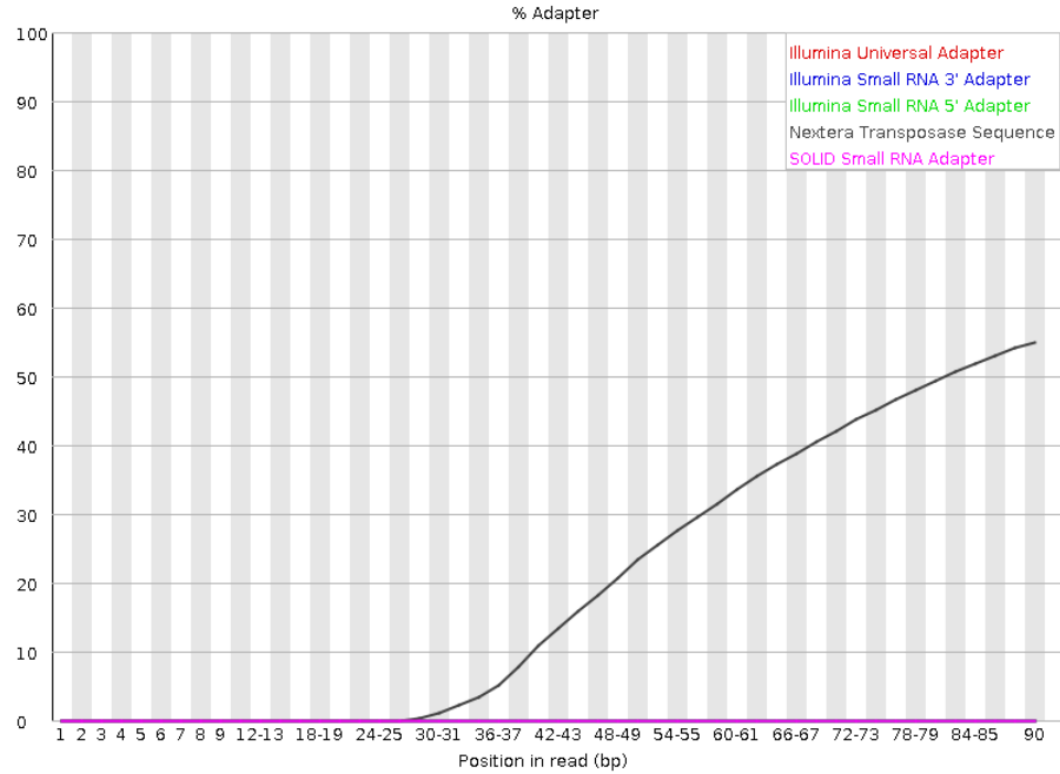
Summary

- ✓ Basic Statistics
- ✓ Per base sequence quality
- ✓ Per tile sequence quality
- ✓ Per sequence quality scores
- ! Per base sequence content
- ✓ Per sequence GC content
- ✓ Per base N content
- ! Sequence Length Distribution
- ✓ Sequence Duplication Levels
- ✓ Overrepresented sequences
- ✗ Adapter Content

✓ Overrepresented sequences

No overrepresented sequences

✗ Adapter Content



A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

Report generated on 2017-09-01, 09:26 based on data in: `/pica/v11/b2017206_nobackup/private/wabi/RNAseq/multiQC_raw_vs_trimmed`

📘 Welcome! Not sure where to start?

Watch a tutorial video

(6:06)

don't show again ✕

General Statistics

📄 Copy table

⚙️ Configure Columns

📊 Plot

Showing 18/18 rows and 4/5 columns.

Sample Name	% Dups	% GC	Length	M Seqs
P8205_101_S1_L001_R1_001	75.7%	46%	126 bp	81.2
P8205_101_S1_L001_R1_001.trimmomatic	75.7%	46%	126 bp	80.5
P8205_101_S1_L001_R1_001.trimmomatic_unpaired	38.7%	47%	106 bp	0.7
P8205_101_S1_L001_R2_001	71.9%	47%	126 bp	81.2
P8205_101_S1_L001_R2_001.trimmomatic	72.0%	46%	126 bp	80.5
P8205_101_S1_L001_R2_001.trimmomatic_unpaired	7.7%	47%	126 bp	0.0
P8205_102_S2_L001_R1_001	83.5%	46%	126 bp	86.2
P8205_102_S2_L001_R1_001.trimmomatic	83.5%	46%	126 bp	85.5
P8205_102_S2_L001_R1_001.trimmomatic_unpaired	50.3%	47%	108 bp	0.7
P8205_102_S2_L001_R2_001	80.0%	47%	126 bp	86.2
P8205_102_S2_L001_R2_001.trimmomatic	80.1%	47%	126 bp	85.5
P8205_102_S2_L001_R2_001.trimmomatic_unpaired	4.0%	41%	126 bp	0.0

FastQC

FastQC is a quality control tool for high throughput sequence data, written by Simon Andrews at the Babraham Institute in Cambridge.

Sequence Quality Histograms

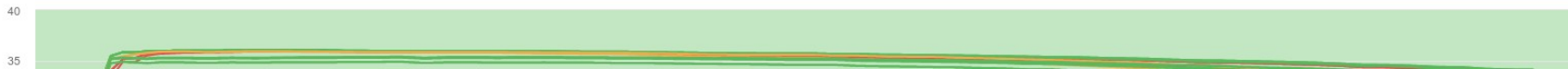
10 2 6

The mean quality value across each base position in the read. See the [FastQC help](#).

Y-Limits: ☒ on

Mean Quality Scores

📄 Export Plot



FastQC is a quality control tool for high throughput sequence data, written by Simon Andrews at the Babraham Institute in Cambridge.

Sequence Quality Histograms 10 2 6

The mean quality value across each base position in the read. See the [FastQC help](#).

Y-Limits: 0 95Per Sequence Quality Scores 15 3

The number of reads with average quality scores. Shows if a subset of reads has poor quality. See the [FastQC help](#).

Y-Limits: 0 95

Per Sequence GC Content

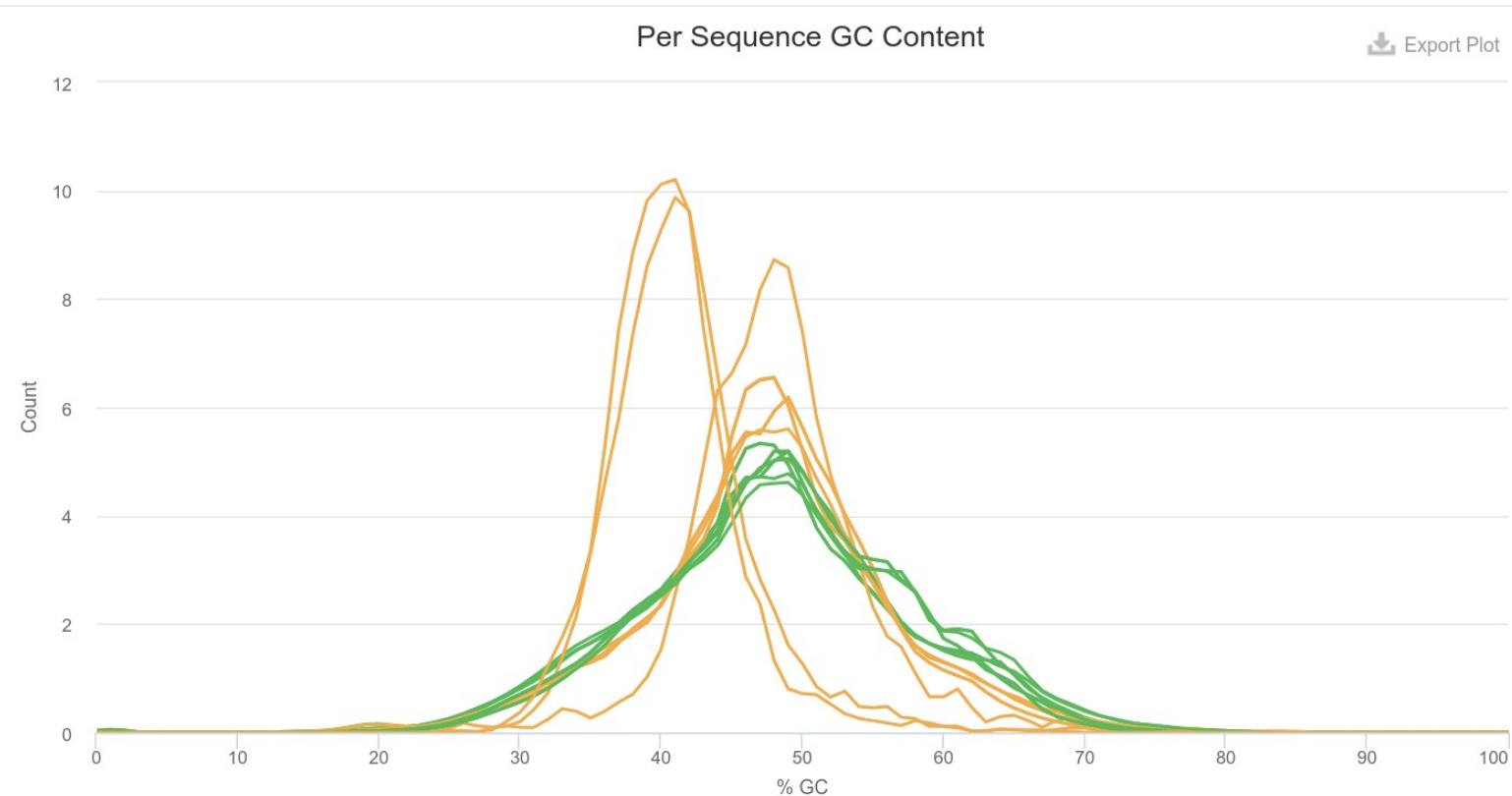
10

8

The average GC content of reads. Normal random library typically have a roughly normal distribution of GC content. See the [FastQC help](#). Y-Limits: ☒ on

Percentages

Counts



The diagram illustrates the Illumina sequencing process, which is divided into four main stages:

- library preparation**: DNA is fragmented and ligated with sequencing adaptors.
- cluster formation**: The library is loaded onto a flow cell, and DNA molecules are amplified to form clusters.
- sequencing**: A laser is used to excite fluorescently labeled nucleotides (ATP, TTP, CTP, GTP) during synthesis.
- sequence identification**: The resulting sequence data is analyzed to identify the original DNA fragments.

A photograph of an Illumina sequencing machine is shown on the right.

- Assigns clusters to a sample, based on the cluster's index sequence which is provided in the sample sheet

Sample_ID	index	index2
Sample1	ATTACTCG	AGGCTATA
Sample2	TCCGAGAG	AGGCTATA
Sample3	ATTACTCG	GCCTCTAT
Sample4	TCCGAGAG	GCCTCTAT

fast

fast

For Research Use Only. Not for use in diagnostic procedures.

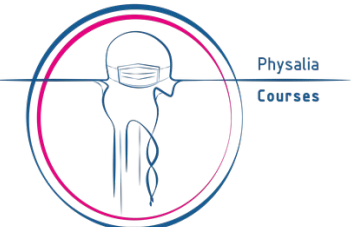


GGTGATACGGCGACCACCACaccgacGGCCCTACACGACGCTCTTCCGATCTXXXXXXAGCACACGTCTGGGCTCCAGTCACgacactaCCGTCTTCTGCTTG

TTACTATGCCGCTGGTGGTgtggctgttGGGATGTGCTGCAGGGGGCTAGAXXXXXXTCGTGTGCAGACTTGAGGTCAGTgctgtgatGGCAGGGGACGGGC

[Adapter/Index Primer] [Index] [Target primer] [Target] [Target primer] [Index] [Adapter/Index Primer]

@K00233:37:HGHLYBBXX:3:1101:2646:1121 1:N:0:NACGCATC+NGCTGGTG
NCGCATGAGCCGCTGTATCAGGCGCTGATCGGGCCGGGCATTGCAGTTGGGATAGATCGGGGGAGCACACGTCTG
+
#A7F<<GG<JFJFJJJJJJJJFJJJJJJJAFFJFJJJJJJJFJAFFJJAJFJJ<FJJJJJFFF<FFA--FFFJJJJJ



Cutadapt

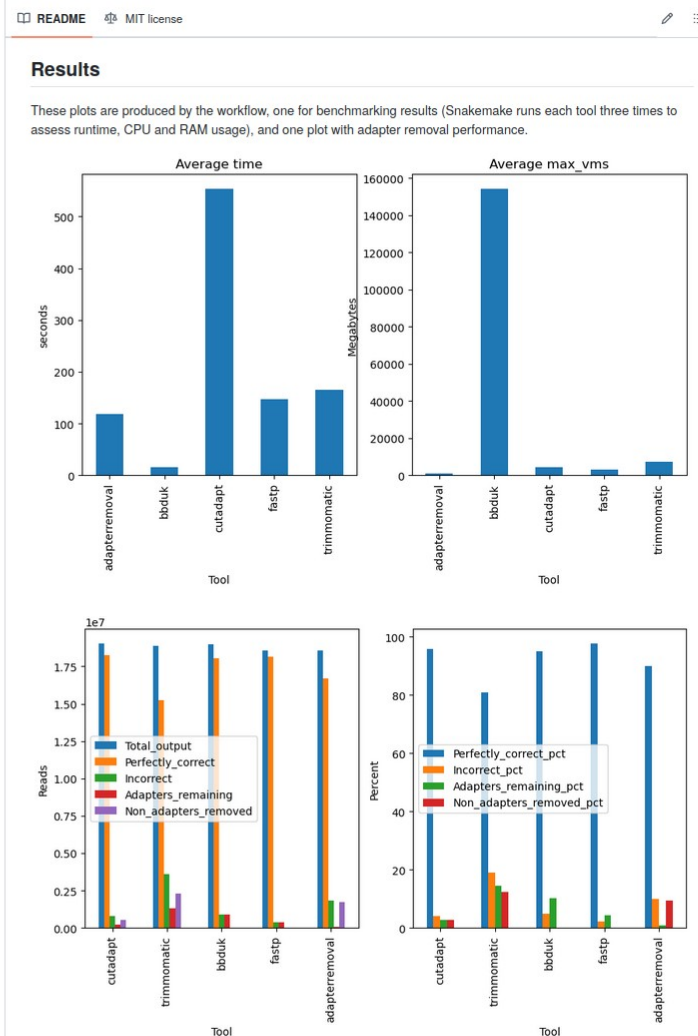
fastp

Trimmomatic

TrimGalore

AdapterRemoval

BBduk



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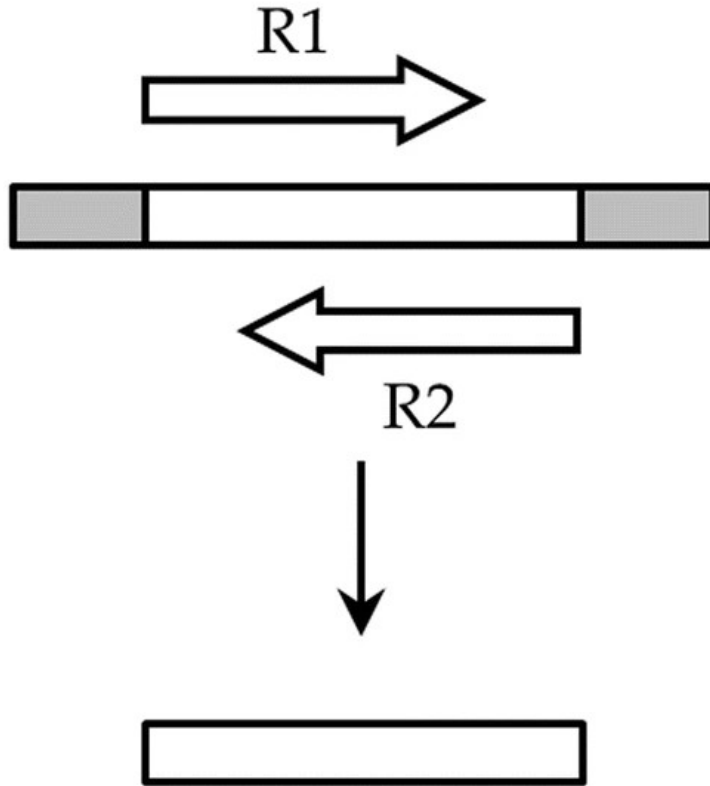
Benchmarking software tools for trimming adapters and merging next-generation sequencing data for ancient DNA

Annette Lien¹, Leonardo Pestana Legori², Louis Kraft¹,
Peter Wad Sackett¹ and Gabriel Renaud^{1*}

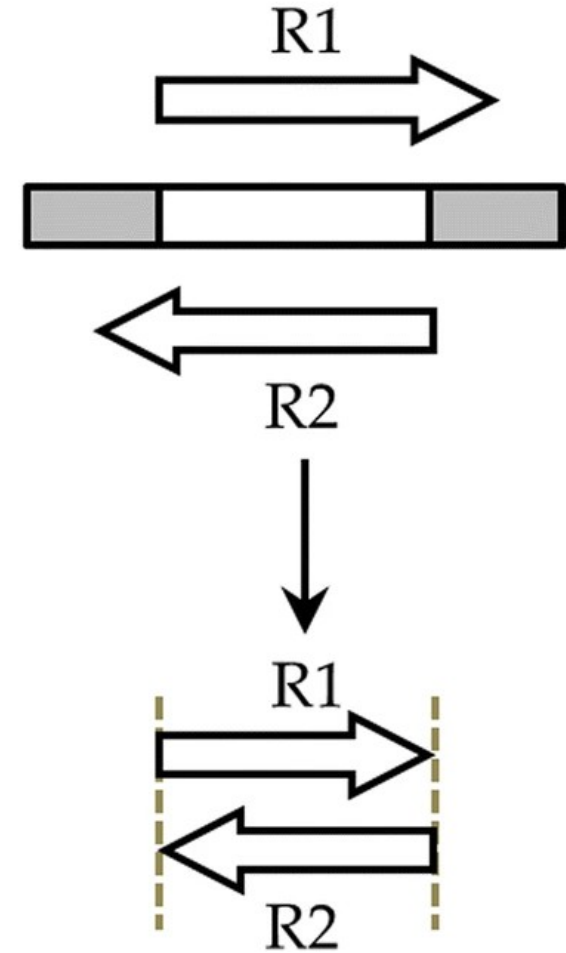
¹Department of Health Technology, Section for Bioinformatics, Technical University of Denmark, Kongens Lyngby, Denmark, ²University of Debrecen, Debrecen, Hungary

Ancient DNA is highly degraded, resulting in very short sequences. Reads generated with modern high-throughput sequencing machines are generally longer than ancient DNA molecules, therefore the reads often contain some portion of the sequencing adaptors. It is crucial to remove those adaptors, as they can interfere with downstream analysis. Furthermore, overlapping portions when DNA has been read forward and backward (paired-end) can be merged to correct sequencing errors and improve read quality. Several tools have been developed for adapter trimming and read merging, however, no one has attempted to evaluate their accuracy and evaluate their potential impact on downstream analyses. Through the simulation of sequencing data, seven commonly used tools were analyzed in their ability to reconstruct ancient DNA sequences through read merging. The analyzed tools exhibit notable differences in their abilities to correct sequence errors and identify the correct read overlap, but the most substantial difference is observed in their ability to calculate quality scores for merged bases. Selecting the most appropriate tool for a given project depends on several factors, although some tools such as fastp have some shortcomings, whereas others like leeHom outperform the other tools in most aspects. While the choice of tool did not result in a measurable difference when analyzing population genetics using principal component analysis, it is important to note that downstream analyses that are sensitive to wrongly merged reads or that rely on quality scores can be significantly impacted by the choice of tool.

A



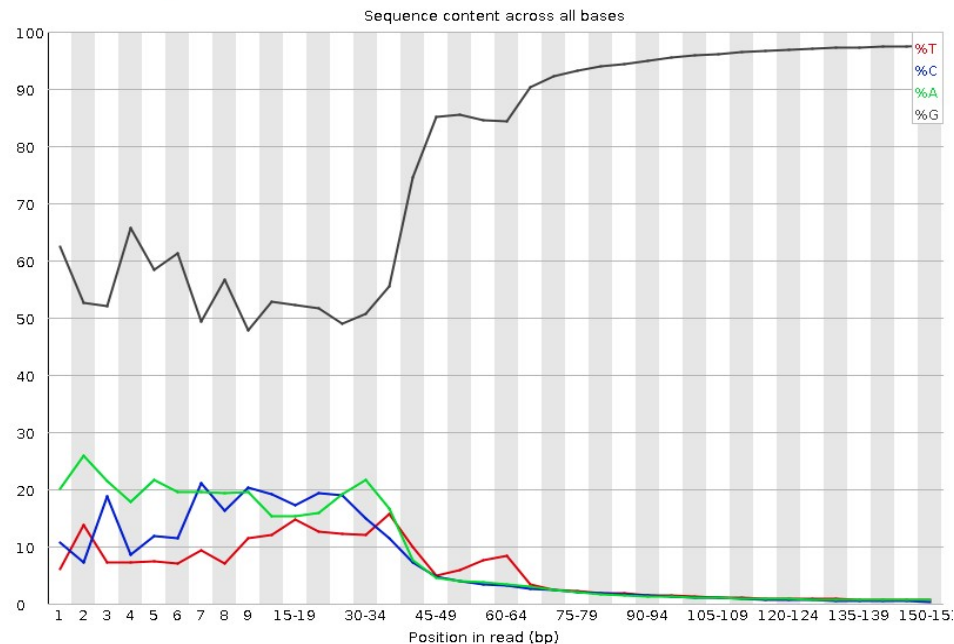
B



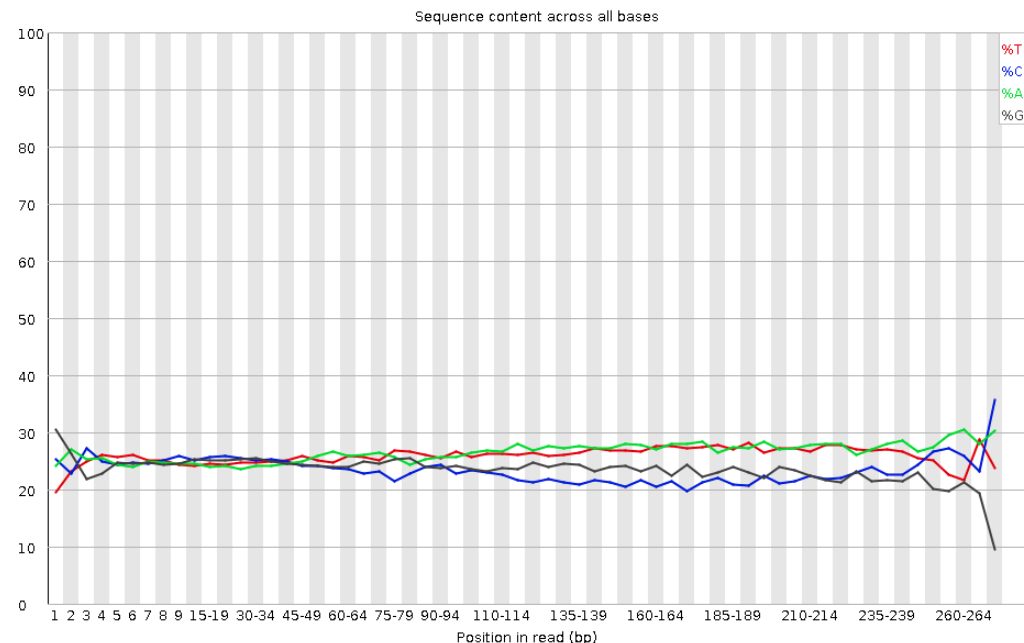
Before fastp

After fastp

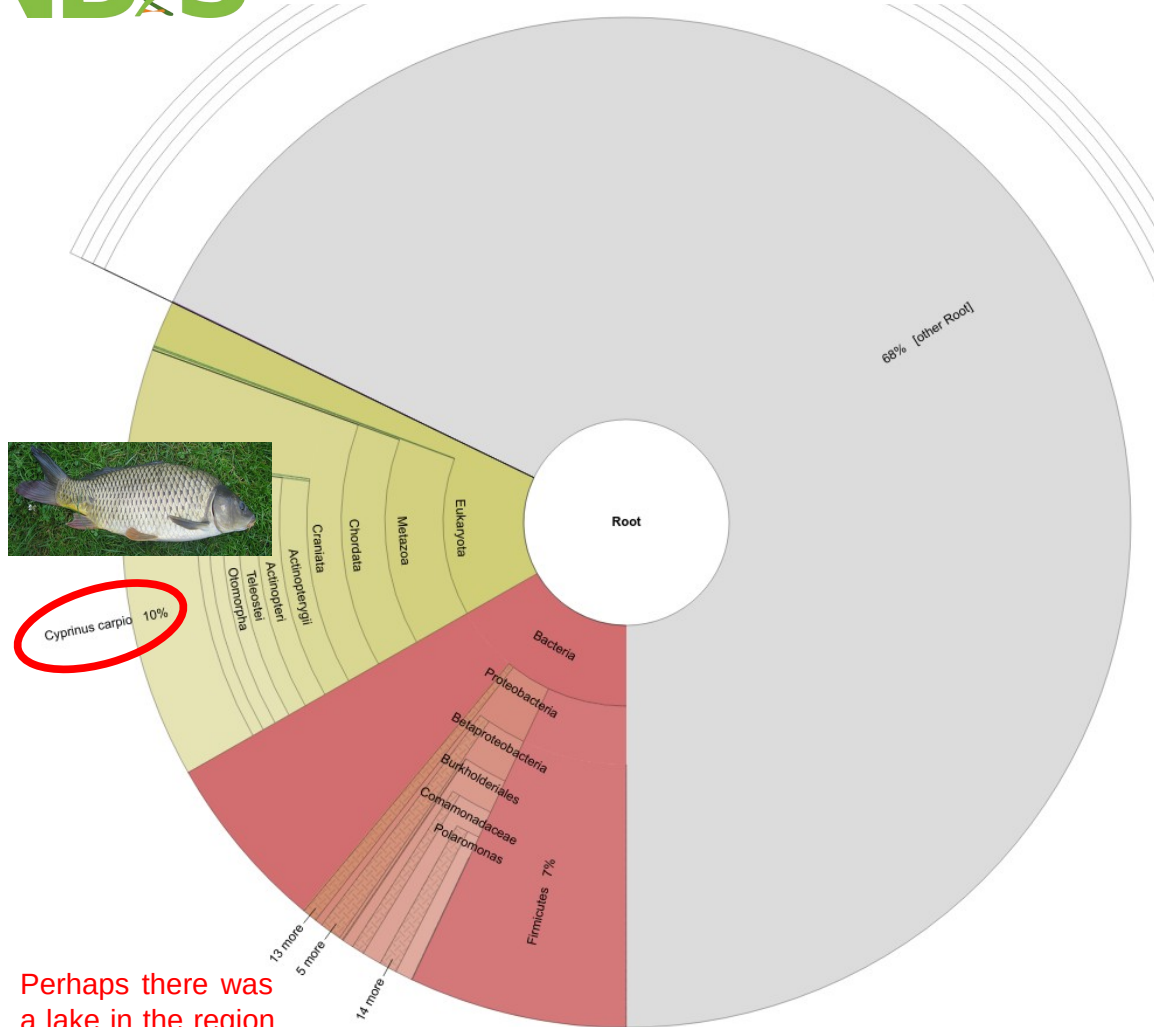
✖ Per base sequence content



✖ Per base sequence content



Was NovaSeq or NextSeq used for sequencing?



Perhaps there was a lake in the region where the soil sample came from?

No, you simply forgot to trim your adapters :)))

It turns out the Carp genome is full of Illumina adapters.

One of the first things we teach people in our [NGS courses](#) is how to remove adapters. It's not hard – we use [CutAdapt](#), but many other tools exist. It's simple, but really important – with De Bruijn graphs you will get paths through the graphs converging on kmers from adapters; and with OLC assemblers you will get spurious overlaps. With gap-fillers, it's possible to fill the gaps with sequences ending in adapters, and this may be what happened in the Carp genome.

Why then are we finding such elementary mistakes in such important papers? Why aren't reviewers picking up on this? It's frustrating.

This is a separate, but related issue, to genomic contamination – [the Wheat genome](#) has PhiX in it; [tons of bacterial genomes do too](#); and [lots of bacterial genes were problematically included in the Tardigrade genome](#) and declared as horizontal gene transfer.

Bioinformatics Bits and Bobs

Rare and random blog posts about bioinformatics, genomics and evolution.

Monday, 29 September 2014

Why you should QC your reads AND your assembly

The genome sequence of the Common Carp *Cyprinus carpio* was [published in Nature last week](#). By coincidence, I was doing some QC on some domesticated Ferret (*Mustela putorius furo*) reads, which had thrown some kmer warnings in the [FastQC tool](#). I blasted the kmers in NCBI and was quite perplexed by the number of hits that I found in the carp genome. Nearly all of the first 150 hits were all from the carp genome. Anyway, I looked a bit further into my odd kmers and it turns out that they were the ends of some Illumina adapter sequences that had presumably been incorporated into the paired-reads on the shorter ends of the insert size. This then took me back to the Carp Genome - what had crept into that?